Julia Allos, Chris Kang
Computational Linguistics
Professor Preum
9 October, 2026

Introduction: Problem and Motivation

Many have heard the joke: A father and his son get into an accident. The father dies at the scene, but the son is raced to the ER. The doctor walks into the operating room and says “I can’t operate on him. That’s my son.” How is this possible? The answer is simple: the doctor is the boy’s mother. Yet the fact that this riddle has remained so effective for decades reveals something less simple about the way we process language. When we hear the word “doctor,” many of us instinctively imagine a man.

This riddle provides a simple illustration of gender stereotyping in language. A person’s interpretation of an otherwise gender-neutral word can be influenced by assumptions about who typically occupies that role. If language models learn from human-generated language, then they may also learn the associations embedded within that language. Can we detect these stereotypes computationally?

Our project explores this question by developing an evaluative model for detecting gender stereotyping in text. We will build a “stereotype judge” that creates a quantitative measurement of gender-stereotype-consistent language. The model will produce an aggregate bias score for a body of text based on the gender stereotypes present in its language, allowing it to be applied to both human- and AI-generated text. For example, the model could be used to compare gender stereotyping in different sources of human writing, evaluate how language models reproduce gender stereotypes, or examine whether different occupations are more strongly associated with one gender than another. The intended users include researchers studying language and social bias and developers evaluating language models.

Dataset selection

This project will use two datasets in training the evaluative model. The first dataset will be CrowS-Pairs (Nangia et al. 2020, EMNLP), a crowdsourced challenge set of 1508 sentence pairs covering

nine categories of United States social bias. It was collected via Amazon Mechanical Turk (MTurk), restricted to workers located in the United States. A crowdworker was given a bias category and asked to write a sentence expressing a stereotype about a disadvantaged group, then edit it minimally so the same sentence describes a contrasting, more advantaged group. This set only has about 300 sentence pairs for gender specifically, however, so there must be another set with it. StereoSet (Nadeem et al. 2020), with 17000 instances, was also collected via MTurk with a US-based worker pool. Workers were given a target term (an occupation, gender term, etc.) and asked to write three completions: one stereotypical, one anti-stereotypical, and one unrelated/nonsensical.

We will alter the dataset so that the unrelated candidate will be dropped, so the dataset will mimic the CrowS-Pairs exactly. This combined set will become the training pool for the classifier. Unlike the CrowS-Pair dataset, whose paired sentences only differ by a word or two, StereoSet's paired sentences vary more in wording and length, so combining the two gives the classifier a wider range of examples to learn and generalize from. For StereoSet's sentences which have a context sentence followed by the actual sentence, we will join these sentences into a single input since the interpretation of the actual sentence depends and changes based on the context. To keep both halves of the data from appearing in the test and training dataset, we will divide the train and test split of data by these data pairs.

Research Question and Baseline Approach

Once the classifier is trained, the research question becomes: Can a classifier trained on human-labeled examples of gender stereotyping learn to detect stereotype-consistent language, and when applied to text generated by a language model, does it reveal measurable, occupation-linked gender bias in that generated text? As a baseline for comparison, we will include a majority-class classifier, which predicts the most frequent label in the training data for every input, independent of sentence content. Because its accuracy is determined entirely by the class distribution of the dataset rather than any learned linguistic signal, this baseline establishes a naive reference point for performance. Because the data is built from pairs, the two classes are roughly balanced by construction, so we expect this baseline to score

close to 50% accuracy. Any proposed classifier must exceed this baseline by a meaningful margin to demonstrate that it is learning linguistic content of the input rather than reflecting label frequency.

Proposed Methods and NLP approach

Our first proposed classifier will use a bag-of-words representation with TF-IDF weighting, followed by logistic regression. Using scikit-learn's TfidfVectorizer, each sentence is converted into a list of numbers, one for every word in the dataset's vocabulary, where each number reflects how important that word is in that particular sentence. We will then train a logistic regression classifier on these numbers using the sentences' human-assigned labels. In practice, this means the classifier starts with a rough guess for how much each word matters, checks how often its predictions are wrong, and adjusts those guesses repeatedly until it settles on a set of weights that best separates stereotype-labeled sentences from non-stereotype ones. After training, we can look directly at the model's learned weights to see which words it associated most strongly with each label.

However, because bag-of-words represents text primarily through individual words rather than their broader context or word order, it may struggle to identify stereotypes that depend on relationships between words or the meaning of an entire sentence. For example, a sentence like "He stayed home to care for the children while his wife went to work" expresses a gendered occupational or domestic-role association without necessarily containing an explicitly stereotypical word. This issue becomes pronounced with the following sentences: "Women are bad at math." "Men are bad at math." Because the label depends on how the gender word combines with the trait, which a linear model over individual words cannot capture, we expect the TF-IDF classifier to perform close to chance on such pairs.

This limitation motivates the use of a more context-sensitive approach based on BERT embeddings, also paired with logistic regression. Unlike a bag-of-words, BERT produces contextualized representations of language, meaning that the representation of a word is influenced by the words surrounding it and the sentence as a whole. We will use a pretrained BERT model from Hugging Face to convert each sentence into a single numerical representation. Each sentence is passed through BERT once,

and we will average its token outputs, mean pool, into one fixed-length vector to represent the whole sentence. We will also test a sentence-transformer model, such as all-mpnet-base-v2, which is trained specifically to produce sentence-level embeddings. These BERT-based representations are then used in place of the TF-IDF ones, with the same logistic regression training process applied on top of them. We will then compare those two approaches. This approach may allow the classifier to identify forms of stereotyping that depend on context, word relationships, or overall sentence meaning rather than individual stereotypical words. If we want to increase the accuracy beyond these logistic regression models, we plan to use small neural networks instead. Unlike logistic regression, neural networks can capture non-linear relationships such as the relationship between a specific trait and gender. To avoid memorizing the training data and overfitting, we plan to use regularization and early stopping. The downside to this approach is that it is not as interpretable as logistic regression.

Evaluation of Dataset and Results

Before training the classifier, we will examine the combined CrowS-Pairs and StereoSet gender dataset once it has been cleaned and merged. First, we will calculate the class balance of this merged dataset rather than relying on class distributions reported in the original papers since this number directly affects the expected accuracy of the majority-class baseline. Second, we will compare sentence length across two labels to check whether stereotype-labeled sentences are systematically longer or shorter than non-stereotype ones. If such a pattern exists, the classifier could end up relying on sentence length rather than actual content. Finally, we will compare word frequencies between two labels to identify which words appear disproportionately in stereotype-labeled sentences. This last step connects directly to this project's central question of whether stereotyping in this data depends mainly on word choice, which the TF-IDF classifier could capture, or on sentence structure and context, which only the BERT-based classifier could capture.

To evaluate the classifiers, we will report standard classification metrics on the held-out test set: accuracy, precision, recall, and F1 score, calculated separately for each label rather than only an overall

average. Beyond these summary metrics, we will separate the mistakes into false positives and false negatives. We will then manually review a sample of these errors and record a likely reason for each misclassification. We expect to find the original human-assigned label being ambiguous or debatable, the sentence depending on subtle or contextual cues rather than explicit trigger words, and unusual sentence structure or length. We will also check whether misclassified sentences disproportionately come from one of the two source datasets, since CrowS-Pairs and StereoSet were constructed differently and may be unevenly difficult for the classifier. Finally, we will directly compare the two classifiers' errors against each other by identifying sentences that the TF-IDF model misclassified but the BERT model classified correctly and vice versa.

Application to Generated Text and Demonstration

To address the second half of our research question, we will apply the trained judge to text generated by a language model. We will select a list of approximately 10–15 occupations spanning traditionally male-coded, female-coded, and gender-neutral roles. For each occupation, we will prompt a language model using a consistent template, such as "Write two sentences about a [occupation]." We will name the model and fix the number of operations per occupation and the temperature hyperparameter. Each generated sentence will then be passed through our chosen classifier. We will aggregate these predictions by occupation to calculate the percentage of stereotype-consistent generations per role. As a point of comparison, we will count the gendered pronouns the model uses for each occupation. This aggregate comparison constitutes our primary test of whether the language model's generated text reflects measurable, occupation-linked gender bias. For the demonstration of this project, we will be creating a single-page app with a textbox where a user types any sentence and submits it to the trained classifier. The classifier will then return the predicted label (stereotype-consistent/not) and a confidence score.

Ethical Considerations and Anticipated Challenges

There are certain ethical considerations to be addressed. For example, defining what is "gender-biased" is difficult as well as which professions are more "difficult" and therefore more "masculine". Additionally, it is important to note that measuring bias this way captures statistical association. It does not capture intent or true sexism in a philosophical sense.

There are also some possible challenges in this project. Merging two separate datasets that are structured differently requires careful harmonization, and inconsistencies introduced during that process could affect downstream results. A further challenge is that a classifier trained on these datasets may end up relying primarily on surface-level word associations (like pairing certain pronouns with certain traits or occupation words) rather than any deeper understanding of context. Rather than treating this as something to avoid, we are treating it as one of the specific questions this project is designed to investigate: comparing the lexical TF-IDF classifier directly against the contextual BERT-based one, and examining where the two disagree during error analysis, will let us assess the extent to which this concern applies to our own results.

Division of Labor

There are two students working on this project, Julia Allos and Christopher Kang. We intend to divide the work equally. Julia will handle the merge of the two datasets and the initial exploration of the data. She will also handle the initial TF-IDF classifier. Chris will then do the BERT classifier and the baseline model. Together, we will compare the results and jointly decide which classifier becomes the official judge. From there, Julia will build the generation pipeline, writing the occupation-based prompt templates and collecting text from the language model, while Chris will build the scoring pipeline, running that generated text through the chosen judge and aggregating the results. We will build the interactive demo together once both pipelines are complete. The final write-up will be divided the same way. Each of us will write the sections describing our own contributions.

AI Statement: I, Julia Allos, used AI to learn more about baseline models and TF-IDF models before writing this proposal. I was not familiar with these topics, so I asked Claude to help me understand what they were and asked it to tell me about different types of baseline models. I prompted Claude on 10/4/26. I, Christopher Kang, used AI to review our proposal. I used it to find potential improvements while at the same time also contributing some of my improvements as well. Specifically, I used the Claude Word add-in. I was able to verify, accept and reject ideas based on the concepts that I learned from other classes in the department as well as being involved in machine learning research. I prompted Claude on 10/04/2026.